
Week 39: Continuous-time Markov Chains

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

1. Theory

Recall that we previously have worked with discrete-time Markov chains, where the stochastic process is indexed by discrete time steps $t = 0, 1, \dots$. We can generalize this to continuous time, meaning the time t can take any value in $[0, \infty)$. Note that we still have a discrete state space $\mathcal{S}$.

Definition 1.1. (*Continuous-time Markov Chain*) A continuous-time Markov chain (CTMC) is a stochastic process $\{X(t) : t \geq 0\}$ with discrete state space $\mathcal{S}$ that satisfies the Markov property and has stationary transition probabilities.

Remarks: For all $s \geq 0$, $t > 0$, and $i, j \in \mathcal{S}$, we have

- Markov property:

$$P(X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j \mid X(s) = i).$$

- Stationary transition probabilities (time homogeneity):

$$P(X(s+t) = j \mid X(s) = i) = P(X(t) = j \mid X(0) = i).$$

Definition 1.2. (*Transition probability functions*) Let $\{X(t) : t \geq 0\}$ be a CTMC with state space $\mathcal{S}$. We call

$$P_{i,j}(t) = P(X(t) = j \mid X(0) = i), \quad t \geq 0, \quad i, j \in \mathcal{S}$$

the transition probability functions.

Theorem 1.3. (*Chapman-Kolmogorov*) For a CTMC,

$$P_{i,j}(s+t) = \sum_{k \in \mathcal{S}} P_{i,k}(s) P_{k,j}(t)$$

for all $t, s \geq 0$ and $i, j \in \mathcal{S}$.

Proof By the law of total probability

$$\begin{aligned} P_{i,j}(s+t) &= P(X(s+t) = j \mid X(0) = i) \\ &= \sum_{k \in \mathcal{S}} P(X(s) = k \mid X(0) = i) P(X(s+t) = j \mid X(s) = k, X(0) = i) \\ &= \sum_{k \in \mathcal{S}} P_{i,k}(s) P_{k,j}(t) \end{aligned}$$

□

Definition 1.4. (*Birth-and-death process*) Let $\{X(t) : t \geq 0\}$ be a CTMC with state space $\mathcal{S} = \{0, 1, 2, \dots\}$ or $\mathcal{S} = \{0, 1, \dots, N\}$. Then $\{X(t) : t \geq 0\}$ is called a birth-and-death process with nonnegative birth rates λ_i and death rates μ_i if $P_{i,j}(t)$ satisfies

1. $P_{i,i+1}(h) = \lambda_i h + o(h)$ as $h \rightarrow 0^+$ whenever $i+1 \in \mathcal{S}$.
2. $P_{i,i-1}(h) = \mu_i h + o(h)$ as $h \rightarrow 0^+$ whenever $i-1 \in \mathcal{S}$.
3. $P_{i,i}(h) = 1 - (\lambda_i + \mu_i)h + o(h)$ as $h \rightarrow 0^+$ for $i \in \mathcal{S}$.
4. $P_{i,j}(h) = o(h)$ as $h \rightarrow 0^+$ whenever $|i-j| > 1$.
5. $P_{i,j}(0) = \delta_{i,j}$.
6. $\mu_0 = 0$ and, for a finite state space, $\lambda_N = 0$.

Remarks:

- At most two possible jumps: $i \rightarrow i+1$ and $i \rightarrow i-1$.
- For the limit used here, $f(h) = o(g(h))$ as $h \rightarrow 0^+$ means that for every $c > 0$, there exists $\delta > 0$ such that

$$|f(h)| < c|g(h)|, \quad \text{for all } 0 < h < \delta.$$

Definition 1.5. (Pure birth and pure death processes)

- Pure birth process: $\mu_i = 0$ for all $i \geq 0$.
- Pure death process: $\lambda_i = 0$ for all $i \geq 0$.

Remark: A Poisson process is a pure birth process with $\lambda_i = \lambda$ for all $i \geq 0$.

Theorem 1.6. In a birth-and-death process with birth rates λ_i and death rates μ_i , the following holds:

- Independent sojourn times.
- $S_i \sim \text{Exp}(\lambda_i + \mu_i)$ for $i \geq 0$.
- Jump probabilities in state i are

$$P(i \rightarrow i-1) = \frac{\mu_i}{\lambda_i + \mu_i} \quad \text{and} \quad P(i \rightarrow i+1) = \frac{\lambda_i}{\lambda_i + \mu_i}.$$

- Here we use the term jump probability from state i to state j to denote the probability that the next transition in state i will be to state j .

Definition 1.7. (Alternative definition of birth-and-death processes) The birth and death process with birth rates λ_i and death rates μ_i can be constructed in the following way.

In each state i , there are two independent competing processes

- $T_1 = \text{Time until birth} \sim \text{Exp}(\lambda_i)$
- $T_2 = \text{Time until death} \sim \text{Exp}(\mu_i)$

If one of the rates is zero, its clock is taken to be ∞ ; that transition cannot occur.

If $T_1 < T_2$, then $i \rightarrow i+1$ (birth); if $T_2 < T_1$, then $i \rightarrow i-1$ (death). Hence,

- If $\lambda_i + \mu_i > 0$, the sojourn time is $S = \min\{T_1, T_2\} \sim \text{Exp}(\lambda_i + \mu_i)$.

If both rates are zero, state i is absorbing and no jump occurs.

Theorem 1.8. If $T_i \sim \text{Exp}(\alpha_i)$ for $i = 1, \dots, n$, and $T_1, \dots, T_n$ are independent, then

$$\min(T_1, \dots, T_n) \sim \text{Exp}\left(\sum_{i=1}^n \alpha_i\right).$$

Theorem 1.9. $T \sim \text{Exp}(\lambda)$ is memoryless, meaning $P(T > s+t \mid T > s) = P(T > t)$ for all $s, t \geq 0$.

Theorem 1.10. (Forward Kolmogorov differential equations) For a birth-and-death process on $\mathcal{S} = \{0, 1, 2, \dots\}$,

$$\begin{aligned} P'_{i,0}(t) &= -\lambda_0 P_{i,0}(t) + \mu_1 P_{i,1}(t), \quad t \geq 0 \\ P'_{i,j}(t) &= \lambda_{j-1} P_{i,j-1}(t) - (\lambda_j + \mu_j) P_{i,j}(t) + \mu_{j+1} P_{i,j+1}(t), \quad t \geq 0, \quad j \geq 1, \end{aligned}$$

with initial conditions $P_{i,j}(0) = \delta_{i,j}$.

For a finite state space $\mathcal{S} = \{0, 1, \dots, N\}$, the equation at the upper boundary is

$$P'_{i,N}(t) = \lambda_{N-1}P_{i,N-1}(t) - \mu_N P_{i,N}(t).$$

Example 1.11. Consider a pure birth process with $\lambda_0 \neq \lambda_1$. Derive $P_{0,1}(t)$

Solution. Using Chapman-Kolmogorov, we have

$$\begin{aligned} P_{0,1}(t+h) &= P_{0,0}(t)P_{0,1}(h) + P_{0,1}(t)P_{1,1}(h) \\ &= P_{0,0}(t)[\lambda_0 h + o(h)] + P_{0,1}(t)[1 - \lambda_1 h + o(h)] \end{aligned}$$

Now, rearranging and dividing by h gives

$$\begin{aligned} P'_{0,1}(t) &= \lim_{h \rightarrow 0^+} \frac{P_{0,1}(t+h) - P_{0,1}(t)}{h} \\ &= \lim_{h \rightarrow 0^+} (P_{0,0}(t)[\lambda_0 + o(1)] + P_{0,1}(t)[- \lambda_1 + o(1)]) \\ &= \lambda_0 P_{0,0}(t) - \lambda_1 P_{0,1}(t), \quad t \geq 0. \end{aligned}$$

We know that $P_{0,1}(0) = 0$, and due to exponential sojourn times that $P_{0,0}(t) = e^{-\lambda_0 t}$. Therefore, we have the differential equation

$$\begin{aligned} P'_{0,1}(t) + \lambda_1 P_{0,1}(t) &= \lambda_0 e^{-\lambda_0 t}, \quad t \geq 0, \\ P_{0,1}(0) &= 0. \end{aligned}$$

We can solve using an integrating factor. Recall that for a general

$$\frac{dy}{dx} + P(x)y = Q(x)$$

we multiply each side by $\mu(x) = \exp(\int P(x) dx)$ and use the product rule. In our case we have the integrating factor $e^{\lambda_1 t}$, and find

$$\begin{aligned} e^{\lambda_1 t} \left(\frac{d}{dt} P_{0,1}(t) + \lambda_1 P_{0,1}(t) \right) &= \lambda_0 e^{\lambda_1 t} e^{-\lambda_0 t} \\ \Rightarrow \frac{d}{dt} (e^{\lambda_1 t} P_{0,1}(t)) &= \lambda_0 e^{(\lambda_1 - \lambda_0)t} \\ \Rightarrow e^{\lambda_1 t} P_{0,1}(t) &= \int_0^t \lambda_0 e^{(\lambda_1 - \lambda_0)\tau} d\tau + C = \frac{\lambda_0}{\lambda_1 - \lambda_0} (e^{(\lambda_1 - \lambda_0)t} - 1) + C \\ \Rightarrow P_{0,1}(t) &= \frac{\lambda_0}{\lambda_1 - \lambda_0} (e^{-\lambda_0 t} - e^{-\lambda_1 t}) + C e^{-\lambda_1 t}. \end{aligned}$$

Using the initial condition $P_{0,1}(0) = 0$ gives $C = 0$.

Problem 1.1. Show that the Poisson process $\{X(t) : t \geq 0\}$ with rate λ satisfies the Markov property.

Solution. Let $N(t) = X(s+t) - X(s)$ be the number of arrivals in the interval $(s, s+t]$. By independent and stationary increments, $N(t)$ is independent of the history up to time s and has a $\text{Poisson}(\lambda t)$ distribution. Therefore, for all $s \geq 0$, $t > 0$ and $i, j \in \mathcal{S}$,

$$\begin{aligned} P(X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) &= P(X(s+t) - X(s) = j - i) \\ &= P(N(t) = j - i) \\ &= P(X(s+t) = j \mid X(s) = i), \end{aligned}$$

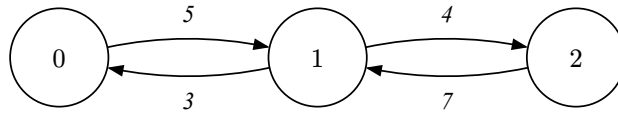
where the probabilities are zero when $j < i$.

Problem 1.2. Consider the birth-and-death process with $\mathcal{S} = \{0, 1, 2\}$, birth rates $\lambda_0 = 5$, $\lambda_1 = 4$ and $\lambda_2 = 0$, and death rates $\mu_1 = 3$ and $\mu_2 = 7$.

1. Draw the transition diagram.
2. Calculate expected sojourn times.
3. Calculate the probability that state 2 follows state 1.
4. Assume $X(0) = 0$. Calculate the probability that two consecutive births occur.

Solution.

1. The transition diagram is



2. We use that the sojourn times are exponentially distributed with rate $\lambda_i + \mu_i$. Thus

$$\mathbb{E}[S_0] = 1/\lambda_0 = 1/5,$$

$$\mathbb{E}[S_1] = 1/(\lambda_1 + \mu_1) = 1/(4 + 3) = 1/7,$$

$$\mathbb{E}[S_2] = 1/\mu_2 = 1/7.$$

3. By the jump probabilities, we have

$$P(1 \rightarrow 2) = \frac{\lambda_1}{\lambda_1 + \mu_1} = \frac{4}{7}.$$

4. Using the Markov property, we have

$$P(0 \rightarrow 1 \rightarrow 2) = P(0 \rightarrow 1)P(1 \rightarrow 2) = \frac{\lambda_0}{\lambda_0 + \mu_0} \cdot \frac{\lambda_1}{\lambda_1 + \mu_1} = \frac{5}{5} \cdot \frac{4}{7} = \frac{4}{7}.$$

Problem 1.3. (Problem 2, Exercise 6) A parking garage opens at 08:00 and closes at 16:00. We model the arrival of cars to the parking garage during opening hours as a Poisson process with rate $\lambda = 0.5$ cars/min. For now, assume that the garage has room for infinitely many cars.

1.
 - Calculate the probability that no car has arrived at 08:05.
 - Compute the expected number of cars that will arrive during the first 15 minutes of opening hours.
 - Given that two cars will arrive during the first 10 minutes, compute the probability that no cars arrive during the first 5 minutes.
2. Customers spend on average 100 kr for parking, independently of each other, with standard deviation of 10 kr.
 - Calculate the expected value of the total income at the garage during one day.
 - Calculate the variance of the total income at the garage during one day.

Assume additionally that cars, independently of each other and their arrival times, spend a stochastic time in the parking garage, which follows an exponential distribution with expectation $1/\mu = 30$ minutes. After the time has passed, the car immediately exits the parking garage. Further, the parking garage has a maximum capacity of $N_{\max} = 20$ cars, and cars that arrive when the garage is full will drive past the garage without forming a queue. Under these assumptions, the number of cars $X(t)$ in the garage at time t is a birth-and-death process.

1.
 - Determine the birth and death rates of the process.
 - Draw a transition diagram for the process.
 - Two cars are in the parking garage at 16:00, and no new cars may enter. The two cars will leave according to the rates described above. Let T be the time until the parking garage is empty, and determine the probability density $f_{T(t)}$.
2. Ignore the opening hours and assume that the parking garage is always open. There are currently 16 cars in the parking garage.
 - Determine the distribution of the time until the number of cars in the garage changes.
 - Calculate the probability that the next time the number of cars in the garage changes, it will be a car that leaves.

Solution.

1.
 - Let $N(t)$ denote the number of cars that have arrived at the parking garage by time t . We see

$$P(N(5) = 0) = e^{-0.5 \cdot 5} \frac{(0.5 \cdot 5)^0}{0!} = e^{-5/2}.$$

- Using the properties of a Poisson process, we have

$$\mathbb{E}[N(15)] = 0.5 \cdot 15 = 7.5.$$

- $$\begin{aligned}
 P(N(5) = 0 \mid N(10) = 2) &= \frac{P(N(10) = 2 \mid N(5) = 0)P(N(5) = 0)}{P(N(10) = 2)} \\
 &= \frac{P(N(10) - N(5) = 2)P(N(5) = 0)}{P(N(10) = 2)} \\
 &= 0.25.
 \end{aligned}$$

2.
 - Let C_i be the amount of money spent by the i -th customer, and X be the total amount paid by all $N(t)$ customers. I.e.

$$X(t) = \sum_{i=1}^{N(t)} C_i.$$

For the entire day (8 hours), the time in minutes is $t = 8 \cdot 60 = 480$ minutes. Using the law of total expectation, we see $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid N(480)]]$. We have

$$\mathbb{E}[X|N] = \mathbb{E}[C_1 + \dots + C_N] = N \cdot \mathbb{E}[C_i] = 100N,$$

and inserting numbers we see

$$\mathbb{E}[X] = \mathbb{E}[100N(480)] = 100 \cdot 480 \cdot \lambda = 24000.$$

- By the law of total variance

$$\text{Var}[X] = \mathbb{E}[\text{Var}[X|N]] + \text{Var}[\mathbb{E}[X|N]].$$

As previously, $\text{Var}[X|N] = N \cdot \text{Var}[C_i] = 10^2 N$, so

$$\mathbb{E}[\text{Var}[X|N]] = 100\mathbb{E}[N] = 100 \cdot 480 \cdot 0.5 = 24000.$$

Since $\mathbb{E}[X|N] = 100N$, we get

$$\text{Var}[\mathbb{E}[X|N]] = 100^2 \text{Var}[N] = 10000 \cdot 480 \cdot 0.5 = 2400000.$$

This gives the total variance $\text{Var}[X] = 2424000\text{kr}^2$.

We now move on to assuming the stochastic parking time

1. • An arrival is accepted at rate λ unless the garage is full. Thus, $\lambda_i = \lambda$ for $i = 0, \dots, N_{\max} - 1$, while $\lambda_{N_{\max}} = 0$. Each car has an independent parking time $U_i \sim \text{Exp}(\mu)$. By the memoryless property, when there are k cars in the garage, the remaining times are independent exponential variables with rate μ . Hence, the time until a car leaves is

$$\min(U_1, \dots, U_k) \sim \text{Exp}\left(\sum_{i=1}^k \mu\right) = \text{Exp}(k\mu).$$

The death rates are $\mu_k = k\mu$ for $k = 1, \dots, N_{\max}$.

- The transition diagram is the linear chain $0, 1, \dots, N_{\max}$, with a transition $k \rightarrow k+1$ at rate λ for $k < N_{\max}$ and a transition $k \rightarrow k-1$ at rate $k\mu$ for $k > 0$.
2. • For a birth-and-death process, the sojourn time in state i is $\text{Exp}(\lambda_i + \mu_i)$. Given that there are 16 cars in the garage, the time until the number of cars changes is therefore

$$S_{16} \sim \text{Exp}(\lambda_{16} + \mu_{16}) = \text{Exp}\left(0.5 + \frac{16}{30}\right) = \text{Exp}\left(\frac{31}{30}\right).$$

- Using the formula for the jump probabilities, we have

$$P(16 \rightarrow 15) = \frac{\mu_{16}}{\lambda_{16} + \mu_{16}} = \frac{16/30}{0.5 + 16/30} = \frac{16}{31}.$$

Problem 1.4. A machine works (1) or is broken (0). Time to breakdown is $\text{Exp}(\mu)$, and time to repair is $\text{Exp}(\lambda)$. All sojourn times are independent. The machine works at time $t = 0$. Calculate the probability that the machine works at time $t = 10$.

Solution. We have

$$X(t) = \begin{cases} 1 & \text{machine works at time } t \\ 0 & \text{machine broken at time } t \end{cases}$$

and the goal is to find $P_{1,1}(10) = P(X(10) = 1 \mid X(0) = 1)$. By the forward Kolmogorov differential equations, we have

$$P'_{1,1}(t) = \lambda P_{1,0}(t) - \mu P_{1,1}(t),$$

and initial conditions $P_{1,1}(0) = 1$. As we only have two states, we also know that $P_{1,0}(t) = 1 - P_{1,1}(t)$, and we can rewrite the differential equation as

$$P'_{1,1}(t) + \mu P_{1,1}(t) = \lambda [1 - P_{1,1}(t)] \iff P'_{1,1}(t) + (\lambda + \mu) P_{1,1}(t) = \lambda$$

Again, applying the integrating factor $e^{(\lambda+\mu)t}$ gives

$$\begin{aligned} e^{(\lambda+\mu)t} \left(\frac{d}{dt} P_{1,1}(t) + (\lambda + \mu) P_{1,1}(t) \right) &= \lambda e^{(\lambda+\mu)t} \\ \implies \frac{d}{dt} (e^{(\lambda+\mu)t} P_{1,1}(t)) &= \lambda e^{(\lambda+\mu)t} \\ \implies e^{(\lambda+\mu)t} P_{1,1}(t) &= \int_0^t \lambda e^{(\lambda+\mu)\tau} d\tau + C = \frac{\lambda}{\lambda + \mu} (e^{(\lambda+\mu)t} - 1) + C \\ \implies P_{1,1}(t) &= \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda+\mu)t}) + C e^{-(\lambda+\mu)t}. \end{aligned}$$

The initial value $P_{1,1}(0) = 1$ gives $C = 1$, meaning

$$P_{1,1}(10) = \frac{\lambda}{\lambda + \mu} (1 - e^{-(\lambda+\mu)10}) + e^{-(\lambda+\mu)10} = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda+\mu)10}.$$

Problem 1.5. (Problem 1, Exercise 6) A pure birth process starting from $X(0) = 0$ has birth parameters $\lambda_0 = 1$, $\lambda_1 = 3$, $\lambda_2 = 2$ and $\lambda_3 = 5$. Let W_1 , W_2 and W_3 be the stochastic times it takes the process to reach states 1, 2 and 3, respectively.

1. Determine the transition probability functions $P_n(t) = P(X(t) = n | X(0) = 0)$ for $n = 0, 1, 2, 3$.
2. Write W_3 as a sum of sojourn times and thereby deduce that the mean time is $\mathbb{E}[W_3] = 11/6$.
3. Determine the mean of $W_1 + W_2 + W_3$.
4. Determine the variance of W_3 .

Solution.

1. By the forward Kolmogorov differential equations, we have

$$\begin{aligned} P'_0(t) &= -\lambda_0 P_0(t), \\ P'_n(t) &= \lambda_{n-1} P_{n-1}(t) - \lambda_n P_n(t), \quad n = 1, 2, 3 \end{aligned}$$

where the initial conditions are $P_0(0) = 1$ and $P_n(0) = 0$ for $n = 1, 2, 3$. For state 0, we have the general solution $P_0(t) = C e^{-\lambda_0 t}$, and using the initial condition gives $P_0(t) = e^{-t}$. For state 1, we have

$$P'_1(t) + \lambda_1 P_1(t) = \lambda_0 P_0(t) = e^{-t}.$$

We can solve this using an integrating factor. In this case, the integrating factor is $e^{\int_0^t \lambda_1 d\tau} = e^{\lambda_1 t}$, and we have

$$\begin{aligned} e^{\lambda_1 t} \left(\frac{d}{dt} P_1(t) + \lambda_1 P_1(t) \right) &= \frac{d}{dt} (e^{\lambda_1 t} P_1(t)) = e^{\lambda_1 t} e^{-t} = e^{(\lambda_1 - 1)t} \\ \Rightarrow e^{\lambda_1 t} P_1(t) &= \int_0^t e^{(\lambda_1 - 1)\tau} d\tau = \frac{1}{\lambda_1 - 1} (e^{(\lambda_1 - 1)t} - 1) \\ \Rightarrow P_1(t) &= \frac{1}{\lambda_1 - 1} (e^{-t} - e^{-\lambda_1 t}) = \frac{1}{2} (e^{-t} - e^{-3t}). \end{aligned}$$

Applying the same method to the remaining equations gives

$$\begin{aligned} P_2(t) &= \frac{3}{2} (e^{-t} - 2e^{-2t} + e^{-3t}), \\ P_3(t) &= \frac{3}{4} e^{-t} - 2e^{-2t} + \frac{3}{2} e^{-3t} - \frac{1}{4} e^{-5t}. \end{aligned}$$

2. Letting S_0 , S_1 and S_2 be the sojourn times, we want to estimate $W_3 = S_0 + S_1 + S_2$. As $S_i \sim \text{Exp}(\lambda_i)$ and using the linearity of the expectation

$$\mathbb{E}[W_3] = \sum_{i=0}^2 \mathbb{E}[S_i] = \frac{1}{\lambda_0} + \frac{1}{\lambda_1} + \frac{1}{\lambda_2} = 1 + \frac{1}{3} + \frac{1}{2} = \frac{11}{6}.$$

3. Since $W_1 = S_0$, $W_2 = S_0 + S_1$, and $W_3 = S_0 + S_1 + S_2$, we have

$$\mathbb{E}[W_1 + W_2 + W_3] = 3\mathbb{E}[S_0] + 2\mathbb{E}[S_1] + \mathbb{E}[S_2] = 3 + \frac{2}{3} + \frac{1}{2} = \frac{25}{6}.$$

4. Since the sojourn times in this pure birth process are independent, we have

$$\text{Var}[W_3] = \sum_{i=0}^2 \text{Var}[S_i] = \frac{1}{\lambda_0^2} + \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2} = 1 + \frac{1}{9} + \frac{1}{4} = \frac{49}{36}.$$

Problem 1.6. (Exam 2025, Problem 3) A student at a lecture is at any time either focusing on the lecture or not. Assume that the expected length of the focused periods is 5 minutes and that the expected length of the unfocused periods is 1 minute.

- 1. Explain that a (simplified) model of the student is as a two-state continuous time Markov chain X with state space $\{0, 1\}$.
- 2. Determine the infinitesimal matrix A of X .
- 3. Assume the student is focused at the start of the lecture. What is the probability of the student being focused 10 minutes later?

Assume that a group of $n = 40$ students is modeled identically as above by $X_1, \dots, X_n$ independent two-state continuous Markov chains.

- 1. Prove that $Z = X_1 + \dots + X_n$ is a Markov chain.
- 2. Determine a differential equation for the probability distribution of Z .
- 3. Assume all students are focused at the beginning of the lecture. What is the expected number of focused students 10 minutes later?

Solution.

- 1. The student is either focused (1) or unfocused (0). We model the focused and unfocused periods as independent exponential sojourn times with means 5 and 1 minutes, respectively. With this modeling assumption, the future evolution depends only on the current state, so X is a two-state CTMC.
- 2. As $\mathbb{E}[S_1] = 5 = 1/\mu_1$ and $\mathbb{E}[S_0] = 1 = 1/\lambda_0$, we have $\mu_1 = 1/5$ and $\lambda_0 = 1$. Therefore,

$$A = \begin{bmatrix} -\lambda_0 & \lambda_0 \\ \mu_1 & -\mu_1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1/5 & -1/5 \end{bmatrix}.$$

- 3. We are interested in $P_{1,1}(10)$. Denote $p(t) = P_{1,1}(t)$. By the forward Kolmogorov differential equations and using $P_{1,0}(t) = 1 - p(t)$, we have

$$p'(t) = \lambda_0(1 - p(t)) - \mu_1 p(t) = 1 - p(t) - \frac{1}{5}p(t) = 1 - \frac{6}{5}p(t)$$

with initial condition $p(0) = 1$. Solving this differential equation and inserting $t = 10$ gives

$$P_{1,1}(10) = \frac{1}{6}e^{-12} + \frac{5}{6} \approx 0.8333.$$

- 1. Since the individual chains are independent, their joint process is a CTMC. When $Z = k$, exactly k students can lose focus, each at rate $1/5$, and $n - k$ students can gain focus, each at rate 1. Thus, the transition rates depend on the joint state only through Z . It follows that Z is a birth-and-death CTMC on $\{0, \dots, n\}$, with birth rates $\lambda_k = n - k$ and death rates $\mu_k = k/5$.
- 2. Let $q_{k(t)} = P(Z(t) = k)$. For $k = 0, \dots, 40$, the forward equations are

$$q'_k(t) = (41 - k)q_{k-1}(t) - \left(40 - k + \frac{k}{5}\right)q_k(t) + \frac{k+1}{5}q_{k+1}(t),$$

where terms with indices outside $\{0, \dots, 40\}$ are omitted. If all students are initially focused, $q_{40}(0) = 1$ and $q_k(0) = 0$ for $k \neq 40$.

- 3. We are interested in $\mathbb{E}[Z(10) \mid X_{i(0)} = 1, i = 1, \dots, 40] = \sum_{i=1}^{40} \mathbb{E}[X_{i(10)} \mid X_{i(0)} = 1]$ by linearity of expectation. As $X_{i(t)} \in \{0, 1\}$, we have

$$\mathbb{E}[X_i(10) \mid X_i(0) = 1] = P(X_i(10) = 1 \mid X_i(0) = 1) = P_{1,1}(10) = \frac{1}{6}e^{-12} + \frac{5}{6}.$$

Therefore

$$\mathbb{E}[Z(10) \mid X_{i(0)} = 1, i = 1, \dots, 40] = 40 \left(\frac{1}{6} e^{-12} + \frac{5}{6} \right) \approx 33.33.$$